

**Vebjørn Erlandsen**

+47 96047010 | vebjorn.erla@gmail.com

Portfolio & Projects: https://v3rlandsen.github.io/portfolio

## EDUCATION

**M.Sc. Cybernetics and Robotics NTNU Trondheim 2026 – 2028 (ongoing)**

Specialization: Robots and Vessel Control

Current coursework: Linear System Theory (state-space models, LQR, Kalman filtering), Nonlinear Control Systems, Algorithms and Data Structures

**B.Sc. Automation and Electronics Design University of Stavanger 2023 – 2026**

Bachelor's thesis: "Simulation-Based Testing Framework for Embedded Systems Using MQTT and Cross-Compiled Code" – Grade A

## SELECTED PROJECTS

- **Helicopter Lab – Control & State Estimation:** Modeled a 3-DOF laboratory helicopter and designed a full multivariable control and estimation system in MATLAB/Simulink. Progressed from basic PD pitch control to LQR with integral action and feedforward, and from a Luenberger observer to a discrete-time Kalman filter. Derived and linearized the equations of motion, implemented the controllers and observers, and validated the complete system on the physical lab helicopter.
- **Bench Power Supply from Converted ATX PSU:** Converted a standard ATX power supply into a four-channel benchtop laboratory supply featuring fixed 3.3 V, 5 V and 12 V rails plus an adjustable 1.5–37.8 V channel with voltage/current metering. Designed the complete enclosure in FreeCAD and 3D-printed it, planned the internal wiring and protection circuitry in KiCad, and assembled the unit with individual fuses, switches, status LEDs and screw-terminal outputs for safe and practical self use.
- **Simulation-Based Testing Framework for Embedded Systems:** Designed and implemented a full simulation-based testing framework for ABB's Virtual IPS embedded controller. Replaced FPGA hardware I/O with Windows shared memory without modifying the controller binary. Built C# MQTT proxy and Python process simulator. Achieved 1.24 ms average round-trip latency and validated closed-loop performance.
- **Distance-Servo System with Custom PCB:** End-to-end design of a distance-servo system: custom PCB design and manufacturing, STM32-based PID control, RS-422 communication between nodes, and a PC-side monitoring GUI.
- **Vision-Guided Pick-and-Place with ABB Industrial Robot:** Developed a complete vision-guided pick-and-place system using an ABB industrial robot. Implemented QR-code detection and localization with OpenCV, performed camera-to-robot coordinate transformation, and integrated the vision pipeline with the robot controller via python and RAPID. Handled practical challenges to achieve reliable autonomous operation.

## TECHNICAL SKILLS

**Electronics & Embedded Systems**

- ESP32, STM32, real-time systems, interrupts, timers, PWM, pulse counters, low-level protocols, PCB design (KiCad), soldering, oscilloscope, logic analyzer, FreeCAD + 3D printing

**Control & Signal Processing**

- PID, state-space modeling, LQR, Kalman filtering, nonlinear control, observers, digital filters, MATLAB/Simulink

**Programming**

- C/C++, Python, C#, MATLAB, JavaScript, RAPID, Structured Text, ESP-idf, STM32CubeIDE

**Robotics & Automation**

- ABB RobotStudio, OpenCV, PLC programming, industrial communication

**Other**

- FreeRTOS, MQTT, Git, technical reporting

## EXPERIENCE

**Power Supply Engineer** – UiS Aerospace student organization (fall 2025)

**Summer Maintenance Worker** – Grimstad Municipality (summer 2022–2024)

**Summer Assistant** – Big Catering og Event (summer 2025)

**Technical Aid Service** - Grimstad Municipality (summer 2026)

## LANGUAGES

Norwegian (Native) • English (Fluent) • German (Basic)